15 Historic figures in genetics

Mendel

You will no doubt have heard of the botanist-monk Gregor Mendel (1822 – 1884) and his famous pea experiments. Mendel's observations of the different traits expressed by pea plants led to recognising that each parent contributed heritable 'units' or 'particles' to their offspring.

Mendel studied seven characteristics of garden peas. Two of the most obvious contrasting characteristics he used were the shape of seeds and the length of stems. He hand-pollinated the pea plants.

Mendel used statistics to help him realise that inherited characteristics are paired units. When he crossed pure breeding tall plants with pure breeding short plants, all of the first generation (F_1) offspring were tall. When the F_1 generation were crossed with each other, the resulting second generation (F_2) had a ratio of three tall plants to every short plant.

1 Define *trait*.

2 Why would Mendel go to the bother of hand-pollinating pea plants when nature would have done it for him?

3 Define *statistics*.

4 Give the genetic definition for *pure breeding*.

Mendel was unable to identify exactly what, or where, those units of inherited characteristics were found but the realisation that the 'units' had to exist was a huge step forward in science. Prior to Mendel introducing this concept, people had to rely on *morphology* and *taxonomy* when trying to work out species relationships. We now recognise Mendel's 'units' as genes, the basic units of inheritance, but it took time to narrow down exactly what genes were and where genes were located.

5 Which branch of science does this define: *'The study of the form and structure of organisms'*?

6 Which branch of science does this define: *'The classification of plants and animals'*?

7 Suggest possible problems faced by using only physical characteristics to predict parent/offspring relationships.

Mendel came up with two main principles. His first principle, the 'law of segregation', stated that somatic (body) cells have pairs of genes (although he didn't use the word 'genes'), but segregate, or separate, when sex cells (eggs or sperm) form. In the process, each member of the pair becomes part of the gamete (sex cell). Mendel's contemporaries believed that physical characteristics 'blended' when they were passed from one generation to the next. Mendel's research showed him that this was not generally true, but that hereditary 'units' were passed on relatively unchanged. This led to his second principle, the 'law of independent assortment', which states that the phenotype (physical characteristic caused by a gene) for any single characteristic is usually not influenced by the phenotype for another characteristic. Since his laws became widely known (many years after his death) they have formed the foundation for modern genetics and heredity.

8 Define *gamete*.

9 Define *phenotype*.

10 In your own words explain what is meant by Mendel's first principle.

11 In your own words explain what is meant by Mendel's second principle.

Research

12 Write a report on the major achievements of *one* of the following pioneers in genetics research:

a Watson & Crick

b T.H. Morgan

c Avery, MacLeod & McCarty

d Hershey & Chase

e Nirenberg, Ochoa & Khorana

f Boyer, Cohen & Berg

g Reginald Crundall Punnett

h Barbara McClintock

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Sutton and Boveri

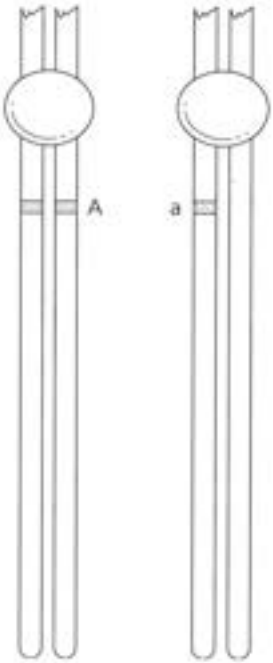
History has many instances of simultaneous discoveries made by scientists working independently. An example occurred in 1902 when American geneticist Walter Sutton (1877–1916) and German scientist Theodor Boveri (1862–1915) both concluded that genes had to be located in the cell's nucleus. Sutton and Boveri reached this conclusion using the logic that to produce a zygote, parents make equal contributions in the form of egg and sperm. Since both contribute equally, whatever the sperm and egg shared in common, in terms of content, had to be where the genes were found. Eggs contain many components absent in sperm. In fact sperm is little more than a nucleus full of genetic material and a tail with which to deliver it. Chromosomes are the main component of the nucleus so the scientists rightly concluded that the genetic material must be located on chromosomes.

Sutton put forward the idea that genes were found on the chromosomes partially as a result of his observations of meiosis.

Morgan

Geneticist Thomas Hunt Morgan (1866–1945) originally doubted the theory that genes were found on chromosomes. He was one of many scientists who believed that genes were separate units and not part of a larger structure. Interestingly, his 1910–1911 research ended up confirming that they were.

Only part of the chromosomes shown



13 Shown here is a simplified diagram of a pair of chromosomes.

On the diagram label the following:

a alleles of a gene

b centromere

c chromatid

14 What does this describe: *'The part of a chromosome to which the spindle attaches during mitosis and meiosis'*?

Decide whether the following statements are true or false:

15 Mendel invented the term *'gene'*. _____

16 A zygote consists of a sperm fused with an egg. _____

17 Genes are found on chromosomes. _____

18 Sperm have the same components as ova. _____

19 Chromosomes are usually found in the cytoplasm of a cell. _____

20 TH Morgan used a variety of fruit fly in his genetics experiments because fruit flies breed very rapidly. _____

21 Sex cells are also called *somatic* cells. _____

22 Meiosis is the process that doubles the number of chromosomes in sex cells. _____

23 Chromosomes are made up of many genes. _____

24 Inherited traits are controlled by factors that occur in pairs. _____

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